Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Multi-agent debate can improve the reliability of large language model answers on scientific questions, but a confident wrong majority can pressure a correct minority agent into abandoning the correct answer. This failure is called sycophantic collapse. Existing mitigations refine prompts or weight agents by self-reported uncertainty; we are not aware of a deployed method that combines claim-level external evidence with a bounded trust state that changes an agent's influence during the debate. This report develops a trust-calibrated deliberation framework in which three heterogeneous agents debate over three rounds, each response is decomposed into atomic claims, and each claim is checked against scientific literature retrieved for its domain. Supported and contradicted claims update a trust vector through logit accumulation, one softmax, and a simplex projection that enforces a floor and a sum-to-one constraint. The final answer is a trust-weighted plurality over the agents' final positions, not a head count. At the mid-project review, the implementation is complete enough to run and test: the P0 suite contains 514 passing tests, the P0 gate reports 13 of 13 checks passed with 0 pending, and Gate 0 completed 10 of 10 vanilla debates with 90 of 90 non-empty positions at a mean of 370.8 seconds per debate. These results validate serving, orchestration, parsing, and the trust operator; they are not an efficacy result. The verifier human audit, the paid Core run, and the Final-stack smoke test remain to be completed, so no claim is made that the framework reduces collapse or improves accuracy.

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Our FYDP instructor Dr. Ohidujjaman set the deadlines and help with the report. Dr. Ohidujjaman's timely responses allowed the team to maintain a consistent pace through out the trimesters.

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Publication List

The publications connected to this research are listed below. One journal article is in preparation; submission is planned after the evaluation. Further submissions are planned once the evaluation results are ready.

Journal Articles

1. Manuscript title (in preparation): The Boundaries of Evidence-Grounded Trust in Multi-Agent Scientific Debate, with a Controlled Benchmark for Sycophantic Consensus

Conference Papers

1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

1.

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Chapter 1

Introduction

1.1 Project Overview

Large language models can answer scientific questions, act as tutors, and support decisions, but they can also state an incorrect answer in fluent and assertive language. Multi-agent debate addresses this by having several agents answer the same question and revise after reading each other's reasoning [1]. Chain-of-thought prompting exposes intermediate steps for the other agents to evaluate [2]. The hard part is aggregating the agents' answers. Most systems use a vote in which every agent has equal weight and the most frequent answer wins [3, 4]. That rule treats repetition as evidence. Scientific inquiry does not. This report examines a trust-calibrated deliberation framework that gives each agent influence according to how well its claims match retrieved scientific evidence.

1.2 Motivation

Debate does not guarantee correctness. Agents can imitate one another or change position under social pressure even when one agent is right. In one study of three-agent debates, 25.5% of the cases in which the agents disagreed had the correct answer in the minority, so majority voting discarded it [5]. Training models to be agreeable can also increase sycophancy and reduce accuracy [6]. Majority voting is vulnerable because a confident majority can pressure a correct minority into switching, and the resulting unanimity hides the error. The risk is highest for scientific questions, where a user may trust a confident but unsupported answer. Existing methods address parts of the problem: iMAD predicts whether a debate is needed [7], DebUnc weights agents by self-reported uncertainty [8], and Mixture-of-Agents combines outputs with static aggregation [9]. None of these methods ties an agent's influence to whether its claims continue to match external evidence during the debate. This study adds that connection.

1.3 Objectives

The objective is to test whether evidence-weighted trust reduces sycophantic collapse relative to majority voting in multi-agent LLM debate. The work has two parts. First, it characterizes the failure: a confident majority can pressure a correct minority into changing its answer [5, 10]. Second, it designs and implements a bounded trust mechanism. Trust rises when retrieved evidence supports an agent's claims and falls when it contradicts them. The trust vector is projected onto a simplex with a floor after every update, so no agent disappears and no agent dominates. The final answer is a trust-weighted plurality, not a head count. The evaluation compares this mechanism with majority voting and other debate and aggregation methods [3, 4]; the planned endpoints are final-answer accuracy, correct-minority retention, collapse under pressure, appropriate revision, and trust calibration.

1.4 Methodology

This study uses an experimental methodology. We first build a vanilla multi-agent debate baseline. Several diverse language-model agents answer the same science question and revise their positions after reading each other's reasoning over three rounds [1]. The baseline aggregates the final positions by majority vote, which exposes the failure modes that the trust mechanism is designed to address [4]. We then decompose each agent's response into claims and check each claim against scientific literature retrieved from external sources [11]. A verifier assigns each claim to supported, contradicted, or unverifiable [12]. Support and contradiction update the agent's trust; unverifiable claims do not move it. Unlike DebUnc, the signal here is external evidence, not self-reported confidence [8]. The trust vector is bounded by the simplex projection described in Chapter 3, and the final answer is a trust-weighted plurality. The models are used inference-only, without fine-tuning. The framework has been exercised end to end on a ten-question GPQA slice (Gate 0), which validates serving, orchestration, and parsing. The comparative evaluation on GPQA-Main and MMLU-Pro is planned; no comparative result is reported here.

1.5 Project Outcome

The project produces a reproducible framework for trust-calibrated scientific deliberation. Agent influence is updated from retrieved evidence while the debate runs, and the final answer comes from evidence-weighted trust rather than a majority count. The project also produces an evaluation protocol. The protocol defines final-answer accuracy, correct-minority retention, collapse under pressure, appropriate revision, and trust calibration, and it specifies a pre-registered matrix. It is designed to detect sycophantic collapse [10] and to track whether correct minority answers survive [5]. A wrong answer can still win. The method records which agents the evidence supported and which it contradicted, so

the decision is auditable.

1.6 Organization of the Report

Chapter 2 covers multi-agent debate and sycophancy and defines the gap this study addresses. Chapter 3 records the requirements, the methodology, the system design, and the project plan. Chapter 4 describes the implementation environment, the pipeline, the P0 hardening changes, and the validation evidence available at mid-project. Chapter 5 covers standards, design constraints, sustainability, and the complex engineering analysis. Chapter 6 closes with a summary, the limits of the evidence, and directions for future work.

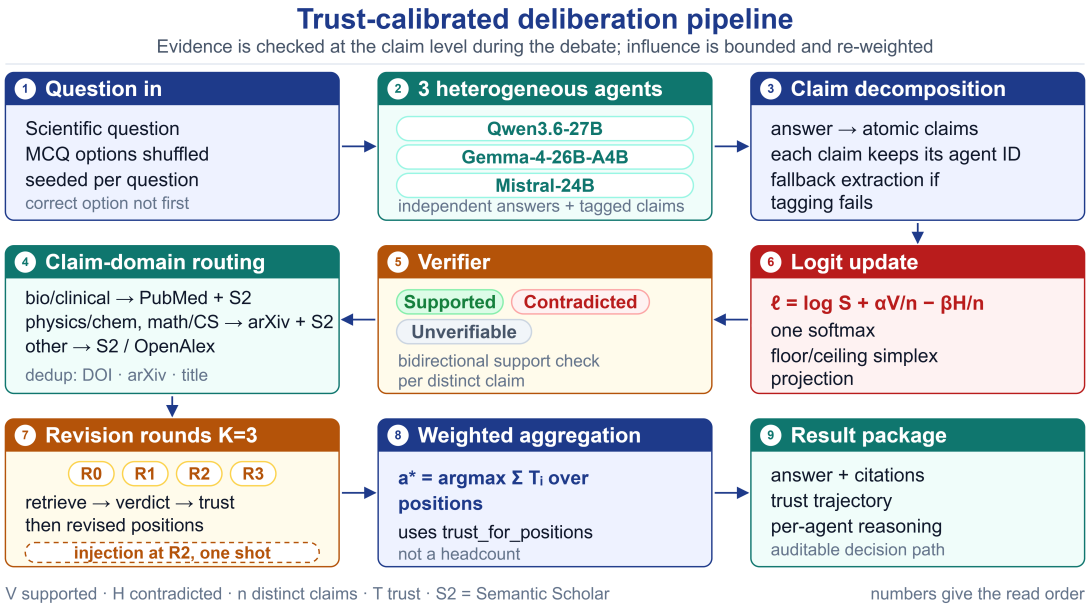


Figure 1.1: End-to-end flow: seeded option rendering, three-round debate, claim decomposition, claim-domain retrieval, verifier labels, logit accumulation with one softmax and a simplex projection, one-shot injection during revision, and trust-weighted aggregation using the final trust vector.

Chapter 2

Background

The chapter introduces the concepts needed to understand the system described in the report and reviews research on multi-agent debate, sycophancy, and agent influence. It then explains the gap addressed by the study.

2.1 Preliminaries

There are several techniques for developing "trust" in an Artificial Intelligence (AI) system. Some of these approaches depend upon a consensus of answers generated by multiple AI systems. Other techniques depend upon finding relevant scientific information and assessing the accuracy of the information found. Chain-of-Thought prompts assist in illustrating the reasoning process that a model uses to arrive at its conclusion, so that we may better understand how an answer was produced and why an error was made during an instance of poor performance [2, 13]. A Multi-Agent Debate Framework provides a methodology where each agent evaluates the arguments presented by the other agents. Each agent presents their proposed solutions and includes support for those proposed solutions. All agents evaluate the proposed solutions and supporting rationales provided by the other agents. Following evaluation and revision of their respective positions and solutions, a final proposed solution is selected [1]. One advantage of allowing each agent to independently evaluate a problem is that if each agent discovers different faults in the system, then collectively the entire team may identify potential problems with the system. Additionally, a team member can utilize another team member's rationalization to criticize aspects of their own rationalization. However, regardless of how well the individual components of the solution are developed, the combination of results (i.e., the methodology of combining results) is equally as important as the results themselves. Most systems combine results with a majority vote. Every agent has an equal say, and the answer supported by the highest number of agents is selected. However, majority voting does not necessarily eliminate errors; rather, it can strengthen repeated errors [3]. Sycophantic Consensus occurs when an agent agrees with the majority position or changes a correct position in favor of a more strongly supported but incorrect position [10]. There are two types of confidence

associated with voting: Self-Confidence in one's position and Token Distribution Confidence. Neither type of confidence verifies the truth claims contained within a position [8]. DebUnc identifies this weakness by comparing uncertainty metrics produced by the deployed system with a Diagnostic Ground Truth Metric that already contains the known answer [8]. Developing trust in an AI system therefore requires obtaining and evaluating scientifically relevant data and evidence.

2.2 Literature Review

The review categorizes the reviewed literature by five categories: debate construction, external verification, sycophancy, uncertainty, and debate evaluation. This study primarily compares methods for determining when to generate a response (i.e., methods for deciding on debate), methods for changing how an agent interacts with another agent (i.e., methods for changing agent interaction), and the method for changing what influences an agent's decision based upon information retrieved from outside the system (i.e., changing agent influence). Recent surveys are provided in [14] as they outline the design space of multi-agent systems utilizing LLMs. In order to develop generated responses that are more reliable, research has shifted focus toward structured reasoning, multi-agent collaboration, and external validation of responses generated by LLMs [13]. Multi-Agent Debate is an example of this new area of research because Multi-Agent Debate allows for other agents to evaluate and potentially challenge each agent's reasoning before that agent produces their final output [1]. The next section outlines some of the most relevant current works in this emerging area and illustrates how the proposed architecture addresses an existing research gap.

2.2.1 Similar Applications

The system is an experimental deliberation layer rather than a conventional web application. It is designed to support scientific question answering, research assistants, and evidence-based decision support. As opposed to the traditional approach where one model solution is generated, the systems listed above generate multiple candidate solutions. Additionally, the systems listed above allow for interaction among candidates. Finally, each system aggregates results from the candidates to generate a final answer for users. Du et al. showed that structured debate among several agents increases factuality and reasoning compared to single-agent approaches [1]. Wang et al. showed that layered aggregation can result in higher overall model capability when combining outputs from several models without supporting explicit debate [9]. Multi-agents systems differ primarily regarding how output is combined. Some systems utilize structured debate as a means to enhance reasoning between agents [1], while others use hierarchical pipelines to aggregate/average the output from each pipeline (the individual agents) [9]. Zhou et al. proposed a self-organizing system for tasks to determine whether a specialized reasoner produces the

response versus being in line with a pre-defined workflow [15]. Additionally, Kushwaha & Madhavan further developed an augmented retrieval framework that identifies contradictions found within retrieved documents and ranks credibility of the source prior to producing the response [12]. Both illustrate how distributed reasoning across specialized agents can improve factual answers. What matters is on what scale you construct dialogue. Ueda et al. analyzed how to construct multi-agent LLM dialogues to facilitate research ideation [16]; however, Wang et al. studied if/how large-scale multi-agent LLMs could provide assistance with scientific code generation [17]. Their aggregation method does not change during dialogue. Similar to a single LLM, the proposed framework has the same reliability goals for answers; yet allows for various levels of influence for each agent throughout the discussion process. After each round of dialogue, evidence verification results are generated; and after each round, the trust scores for all agents are updated based on claim-level scientific evidence supporting their claims. Therefore, an unsupported majority consensus would hold less value than a well-supported minority view.

2.2.2 Related Research

The work of Wei et al. (2022) shows that "Chain-of-Thought Prompting" improves the ability of a large language model to reason [2]. The work of Du et al. (2024) extended the work of Wei et al. (2022) by incorporating multi-agent debate in addition to chain-of-thought prompting, in which multiple agents give answers to a question, look at how other agents have reasoned when answering the question, and then adjust their reasoning [1]. In contrast to both previous approaches for enhancing the performance of large language models, CRITIC [18] relies on outside tools to validate the answers provided and enhance them. It therefore permits validation of assertions contained in each response; however, there is no mechanism to compare additional models working on the same question simultaneously. Instead, Mixture-of-Agents develops numerous independent agent-proposer models that produce answers to identical questions and layers the responses into one output, indicating what several large language models can do together; however, the layering of the individual agent responses is fixed and contains no evidence-based trust values [9]. iMAD primarily targets cost; it also reports accuracy gains in some settings [7]. Fan, Yoon, and Ji identified 41 properties of a structurally defined self-critic and constructed a classifier that determines whether or not to initiate a debate based on those properties. They were able to attain token reductions of up to 92% relative to GroupDebate and increases in accuracy of up to 13.5% over a single-agent baseline. Agent influence remains constant once the debate starts. Pitre, Ramakrishnan, and Wang investigated sycophancy in debates [10]. The authors report that the correct answer was present but ignored in more than 20% of the wrong-answer cases they examined. CONSENSAGENT identifies stalling and answer copying in a debate. When either condition is identified, CONSENSAGENT alters the task prompt before initiating the next stage of the debate. The results of the study demonstrate that measured sycophancy decreased as a result; however, the

total trust score is still dependent upon self-reported confidence and consistency. DebUnc permits agents to express uncertainty among themselves [8]. However, agents express uncertainty via prompts or attention scaling. The deployed measures provided limited benefit in their evaluation. A diagnostic ground-truth signal is available for comparison purposes. The diagnostic ground-truth signal has access to information concerning the correct answer and has greater potential than the deployed measures. The difference between the deployed measures and the diagnostic ground-truth signal indicates that the major barrier is the quality of the trust signal. Kushwaha and Madhavan created an 8-agent MAS-RAG system intended to identify contradictions between retrieved documents and rank evidence based upon credibility [12]. Although source credibility was evaluated by humans prior to deliberation, it was never changed during deliberation. There have also been numerous studies evaluating the social aspects of debate. The authors of Cau et al. identified how selective persuasion could cause an increase in opinion convergence [19]. Yeow et al. studied decision architectures and found that feedback between agents can spread early errors [20]. Wang et al. demonstrated how a good example given in context within one large language model (LLM) may perform better than multi-agent dialogue in certain situations. As noted in [3], judge error and peer conformity were identified as potential drawbacks. The authors of Choi, Zhu, and Li separated communication used in debate from the act of voting and found that majority voting explained a substantial portion of the normal performance gains [4]. FREE-MAD extended previous research by eliminating the need for final agreement and utilizing anti-conformity prompts and trajectory scores [21]. ReSo utilizes dynamic task graphs with reward-based processes for choosing agents [15]. Past research has explored coordination and agent selection. None of these studies combines claim-level scientific evidence with an explicit bounded in-debate trust state and a controlled wrong-majority benchmark. Sycophancy also occurs at the model level. The results of training models to be perceived as “warm” and “agreeable,” as demonstrated by Ibrahim et al., produced lower levels of factual accuracy and higher levels of sycophancy [6]. Chen et al. utilized self-augmented preference alignment from a training perspective to address the issue [22]. Process-level diagnostics for engagement, responsiveness, influence asymmetry, balance, stability, and agent utility developed by Pitre et al. would be helpful when ground truth is unavailable [23]. Final agreement is insufficient to measure a debate, as stated by Pitre et al. [23].

2.3 Gap Analysis

There are still three open issues with regard to Multi-Agent Debate Systems (MADS) from our review of the current literature. The first issue is whether the extra computational burden associated with a MADS can be justified. Fan et al. have used iMAD for this purpose – to predict whether a debate is needed prior to the start of a debate [7]. The second issue is how to detect unsupported agreement caused by conformity or sycophancy. CONSENSAGENT addresses this problem through prompt optimization during

the debate [10]. The third issue is how to determine which agents should receive greater influence when they disagree with one another. DebUnc has a weighting mechanism to deal with uncertainty as it relates to the debate process; however, the trust mechanisms of this weighting rely on each individual agent's internal assessment of uncertainty and therefore do not use external validation or independently verifiable evidence [8]. As previously mentioned, even when multiple agents agree with respect to a particular piece of information, that information may be incorrect. For example, Cau et al. showed that selective persuasion can affect the level of consensus reached by different agents [19], while Wang et al. showed that peer conformity can limit the effectiveness of multi-agent debate [3]. Retrieval-based systems verify retrieved information or improve aggregation, but they do not update an agent's influence from verified scientific evidence during a debate [18]. Similarly, although dynamic coordination mechanisms help solve the issue of selecting which agents participate in the debate, they do not provide a mechanism for increasing or decreasing an agent's influence based on evidence of its behavior during the debate [15]. We are not aware of a method that continuously updates an agent's influence from independently verified evidence during a debate. The proposed framework is designed to address this gap by decomposing each agent's responses into individual factual claims and identifying scientific passages that relate to those claims. Each claim is then classified as either supported, contradicted, or unverifiable. Each verdict feeds an update function for a bounded trust vector. As a result, well-supported minority agents can retain influence during a debate, while an unsupported majority consensus is given less weight. The evaluation protocol defines final-answer accuracy and correct-minority retention as primary endpoints, with collapse under pressure, appropriate revision, and trust calibration as secondary and mechanism measures.

2.4 Summary

The chapter introduced LLM reasoning, chain-of-thought prompting, multi-agent debate, and sycophantic consensus. The reviewed studies cover debate efficiency, prompt correction, uncertainty weighting, external verification, process evaluation, and agent coordination. They do not continuously update agent influence from claim-level scientific evidence. The next chapter uses this gap to define the system requirements, detailed methodology, and project plan.

Chapter 3

Project Design

Chapter 3 defines the system requirements and lays out the methodology, the design decisions, and the project plan with the task allocation. The first section records the functional and nonfunctional requirements together with the context diagram, the level-one data flow diagram, and the interface design.

3.1 Requirement Analysis

Requirement analysis fixes what the framework must do before any code is written. The system is an experimental deliberation layer rather than a conventional application, so the requirements are framed around the research pipeline. A question enters, agents debate it over a fixed number of rounds, the claims are checked against external evidence, and a trust-weighted answer comes out. A multi-agent debate approach is used here since prior research indicates it yields more factual responses than relying on a single model [1]. Furthermore, we incorporated a trust mechanism to prevent situations where a confident but incorrect majority forces a correct minority agent to change its answer [10]. The below sections will provide detail on the individual components of the System Requirements for this project including Boundaries, Data Flows and the User Interface for the Framework.

3.1.1 Functional and Nonfunctional Requirements

The Functional Requirements outline the primary functions that the framework provides. All input questions (from both the User and/or the Evaluation Harness) are processed in complete ignorance of their actual "ground-truth" as it relates to the Agents. In order to be efficient, all input questions do not require a full debate. Therefore, an initial Confidence Gate determines whether an input question should receive a Direct Response or be sent to the Main Pipeline for processing. The Orchestration Process controls 3 different Heterogeneous LLM Agents, each revising at most 3 times per Agent (via State Machine), as determined by 3 retry limits.

Each Agent's Reasoning is segmented into Atomic Factual Claims and carefully tagged to its Source. If this structured tagging process fails, a fallback extraction pass is triggered

to handle the unparsed text. Retrieval routes each claim by domain: biomedical and clinical claims go to PubMed and Semantic Scholar, physics, chemistry, mathematics, and computer-science claims go to arXiv and Semantic Scholar, and other claims go to Semantic Scholar or OpenAlex. Passages are deduplicated by DOI, arXiv identifier, or title, and a cross-encoder reranks the candidates that come back. Each claim then receives one of three verdicts: supported, contradicted, or unverifiable. Contested evidence is reported separately rather than counted as support or contradiction.

After every evidence round the trust vector is updated in the log domain. For agent i in round t, let V_i be the number of supported claims, H_i the number of contradicted claims, and n_i the number of claims checked. The score is

$$\ell_i^{(t+1)} = \log w_i^{(t)} + \alpha \frac{V_i}{n_i} - \beta \frac{H_i}{n_i}, \tag{3.1}$$

where $w_i^{(t)}$ is the current trust weight. One softmax turns the scores into weights, and a simplex projection enforces $w_i \geq f$, $\sum_i w_i = 1$, and $w_i \leq \max(c, 1/N)$. The defaults are $\alpha = 0.5$, $\beta = 0.3$, $f = 0.1$, and $c = 0.9$.

The projection matters. Without it an agent could be silenced outright or could grow to dominate the debate. A clamp followed by renormalization is not equivalent: the old order could leave a weight below the floor, so the implementation projects after a single softmax and tests the result. An injection point lets the harness deliver a fabricated wrong consensus to the targeted agents once, between rounds. That is how sycophantic collapse is measured under controlled pressure. At the end, a weighted plurality over the observed answers selects the final answer, using the trust vector computed from the final positions (`trust_for_positions`). Majority voting is retained as a baseline and logged for comparison; it does not choose the treatment's answer. The results package contains the final answer, the retrieved passages and per-claim verdicts, the trust history, and the transcript for each agent.

The evaluation harness will compute an estimate of the answer's correctness along with a measure of how well the framework has collapsed into a consensus, preserved minority opinions, and calibrated its use of evidence. It will evaluate this framework with the four executable arms; MoA [9], iMAD [7], and DebUnc [8] remain related baselines and are not part of the current matrix. There are several built-in recovery routes in case there is some kind of failure within the system. For example, if a model's output cannot be processed then it will be sent through the fallback extraction path. Any timeouts encountered during this process are retried up to three attempts with exponential backoff before the round is declared inconclusive. Additionally, if a retrieval API becomes unavailable then the system will either default to using available cached evidence or make queries directly to OpenAlex.

With respect to non-functional requirements, which refer to properties of systems, such as performance and scalability, but do not describe specific behavior, latency is currently the most significant limiting factor. This is achieved by forcing adherence to the limit of

three debate rounds plus three retries. This ensures that total processing times remain reasonable on the intended hardware. The system was also designed such that when a single component fails, it does not bring down the entire debate. Instead, failure is dealt with at a local level while keeping other components active. Portability is another important design consideration. By utilizing the vLLM interface compatible with OpenAI, the framework supports both closed APIs and locally hosted models with equal ease. For practical purposes, the same pipeline can operate unmodified on both our development models (Qwen3.5-9B, Gemma 4 12B, Ministral-3-14B), as well as the final evaluation models (Qwen3.6-27B, Gemma 4 26B, Mistral Small 3.2 24B). No fine tuning is required for inference. The pinned final checkpoints total 69.73GB, so the three models share one 96GB RTX PRO 6000 Blackwell GPU with headroom for the KV cache. Using this GPU costs approximately USD 1.10 to 2.70 per hour, with a median near USD 1.60. The pinned final checkpoints total 69.73GB, which leaves headroom on the 96GB card for the KV cache. Finally, each run records its seed, the order of multiple-choice options is derived from a per-question seed, and all relevant intermediate outputs are logged. This allows the final output to maintain an auditable decision trail detailing which agent's claims were supported or contradicted by the evidence. Throughout this process, the ground truth remains completely hidden from the agents, and the final record links supported claims to the passages that supported them.

3.1.2 Context Diagram

As illustrated in Figure 3.1, the context diagram outlines the system boundaries and identifies the external actors interacting with the framework. The core system is positioned at the center, receiving input questions either directly from a user or via the evaluation harness. Scientific passages come back from three external databases, and off-the-shelf language models are called through the vLLM server on a rented RTX PRO 6000 Blackwell GPU. The evaluation harness receives the scores, and the user gets the final result package with the answer, the citations, the trust trajectory, and each reasoning trace.

3.1.3 Data Flow Diagram Level 1

The level-one data flow diagram in Figure 3.2 decomposes the framework into seven processes. A question enters through the intake and gate process, which decides between a direct answer and a debate; for a debate, the option renderer applies a seeded permutation and stores the rendered order. Orchestration runs the revision rounds. It sends inference requests to the external models and reads and writes the debate state. Each position is decomposed into atomic claims, and each claim is routed by domain to PubMed, arXiv, Semantic Scholar, or OpenAlex. The verifier assigns supported, contradicted, or unverifiable, and the verdicts drive the trust update through logit accumulation, one softmax, and a simplex projection. The new trust values are written back into the debate state. One-shot injection is delivered before round two. At the last round, weighted aggregation

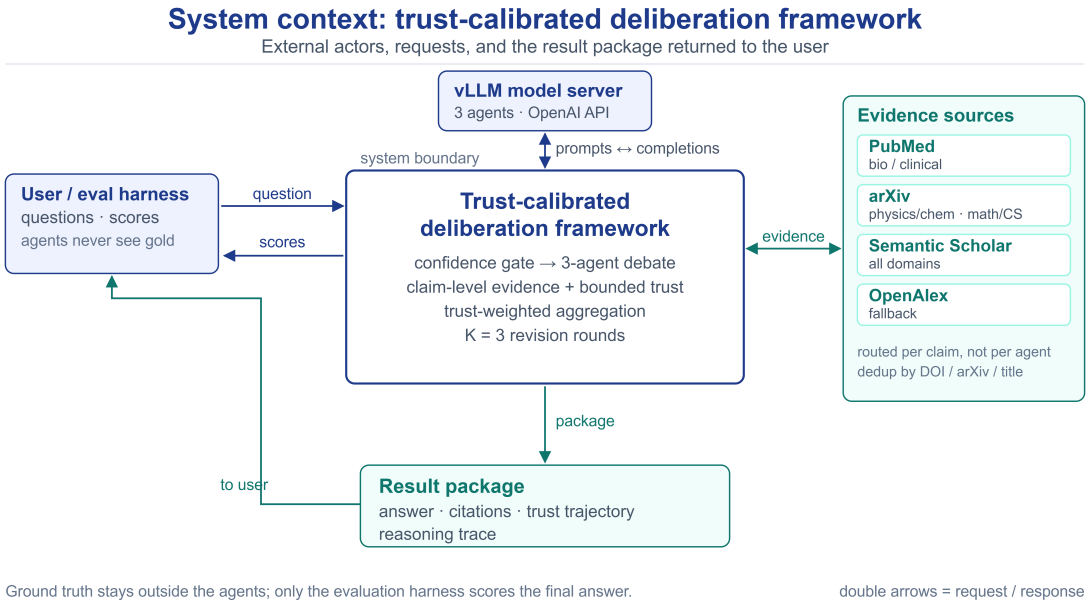


Figure 3.1: Context diagram: questions and rendered options enter; three agents call the vLLM server; claims route by domain to PubMed, arXiv, Semantic Scholar, and OpenAlex; verifier labels update the trust vector; the result package records the final answer, evidence, and final trust vector.

uses the final trust vector and the packaging process logs the result before it returns to the user. Three data stores hold the debate state, the evidence and verdict log, and the results log.

The diagram reads from left to right and top to bottom. The evaluation harness submits a question and its rendered options, and the confidence gate decides whether the debate is worth running. The orchestration process then asks the model server for the agent positions. Positions are decomposed into atomic claims, and the routing process sends each claim to its domain-specific sources. The verifier labels are stored in the evidence log before they drive the trust update. The updated trust is written back into the debate state, so the next round starts from the corrected influence values. At the last round the aggregation process selects the answer with the trust vector computed from the final positions. The packaging process logs it in the results log, and the result package with the answer, evidence, and trust history returns to the user.

3.1.4 UI Design

The framework is operated through a command-line interface. It starts experiments and points the pipeline at the evaluation datasets; every run writes a JSON result package. The command line is the primary interface because the system runs in batches on a rented GPU, where a graphical interface adds little value. A web dashboard for inspecting single debates is planned but is not part of the P0 implementation. Figures 3.3 and 3.4 are layout mock-ups, not screenshots of a running system. The dashboard mock-up shows the

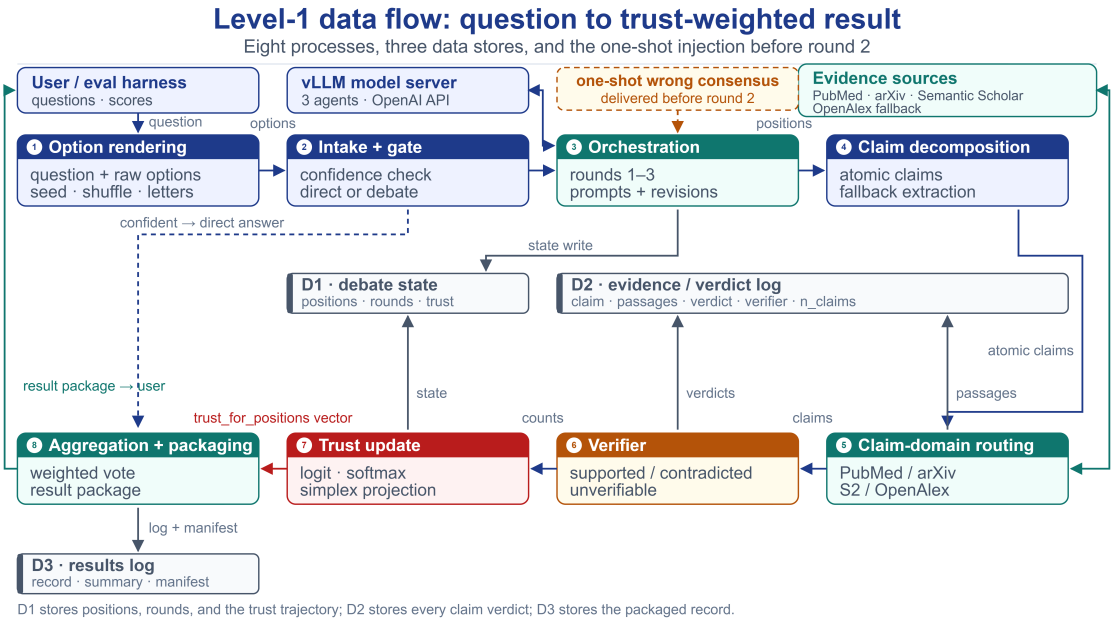


Figure 3.2: Level-one data flow: option rendering, debate state, claim decomposition, domain routing, verifier labels, simplex trust update, one-shot injection, and weighted aggregation.

intended agent cards, trust trajectory, and per-claim verdict view.

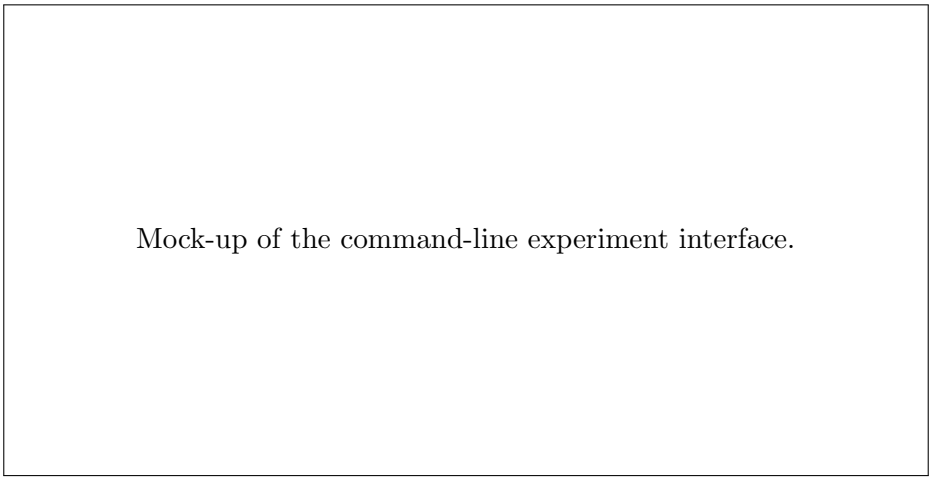


Figure 3.3: Mock-up of the command-line interface for running experiments and writing JSON result packages.

3.2 Detailed Methodology and Design

The design separates the debate from the evidence layer. The debate fixes the number of agents at $N = 3$ and the number of rounds at $K = 3$. The first round collects independent positions; rounds two and three expose each agent to the other agents' positions and to its own current trust weight. The evidence layer runs after each round's positions are fixed. It

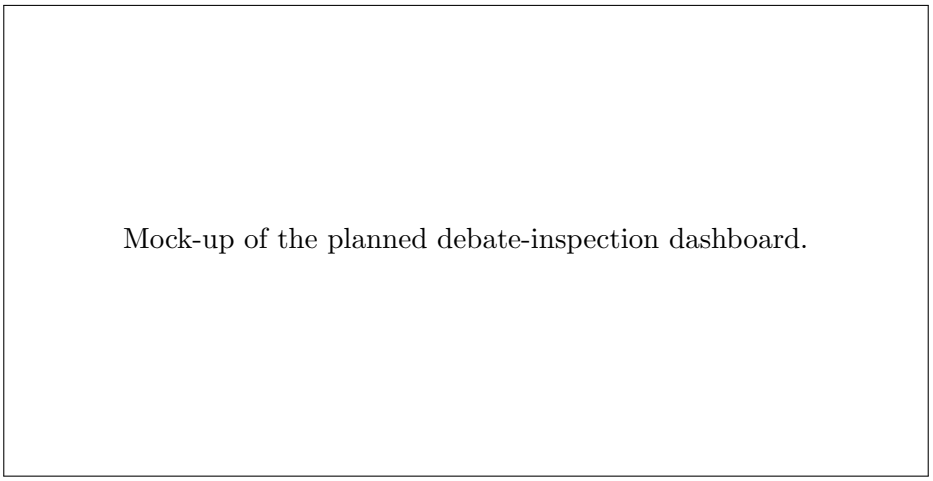


Figure 3.4: Mock-up of the planned web dashboard showing agent positions, the trust trajectory, and per-claim evidence verdicts.

decomposes each position into at most eight claims, routes each claim to a domain-specific source pair, retrieves and deduplicates passages, and assigns a verdict. The verdicts are counted per agent as V_i and H_i, and the trust operator in Section 3.1 updates the vector.

The aggregation step is a weighted plurality. For each distinct final answer a, the score is $\sum_{i:\, p_i=a} w_i$, and the answer with the largest score wins; ties resolve to the first observed answer. The majority answer is computed and stored for comparison, but the treatment's final answer is the weighted winner. This separation lets the same debate transcript be scored under majority and trust-weighted aggregation without re-running the models.

The design also fixes the evaluation contract. Each question is paired across arms by a frozen round-0 prefix and the same option permutation. The primary endpoint is final-answer accuracy; the co-primary endpoint is correct-minority retention. Collapse under pressure, appropriate revision, and trust calibration are secondary or mechanism measures. The pre-registered matrix is described in Section 3.3.

3.3 Project Plan

The project runs for ten months, from Month 1 to Month 10, and covers 40 weeks. It is split into eight phases with six review gates. The work starts with literature and setup and ends with writing, the final defence, and paper submission. Table 3.1 lists each phase with its weeks and deliverables. Table 3.2 lists the gates and their checkpoints. Table 3.3 adds a SWOT view of the plan. This report covers work through the mid-project review in Month 5, while later phases carry the project to the final defence.

The plan starts with scoping. Project scoping in Phase-0 fixes the research question, the trust-calibrated idea, the evaluation shape, and the gate criteria. Data and model work runs in parallel. Datasets GPQA [24] and MMLU-Pro [25] are checked, and the three development models are chosen with their serving setup. Analysis of the baseline follows in Phase-1. Injection tests and the B3 and B4-matched arms are run, and the

pilot checks if a wrong majority can still fool voting. Design and implement fill Phase-2. Claim decomposition and claim-domain retrieval are created. The trust operator uses logit accumulation, one softmax, and a simplex projection. A short dry-run and a design freeze mark the end of Phase-2 to 3. Phase-3a holds the main studies: the core conditions on the primary datasets, with two seeds and 95% confidence intervals. Phase-3b contains ablation studies implementing different agent numbers as well as different alpha/beta combinations. The human study and failure reviews occur in Phase-4. Finally, writing, reproducibility packaging, and defence preparation complete the project in Phase-5, with the paper submission planned for August–December 2027.

Table 3.1: Project phases with week-level durations and deliverables.

Phase	Duration	Deliverables
Phase-0: Literature and Setup	Month 1 (Wk 1–4)	Literature freeze, vLLM serving and Python debate-loop setup, model identification, vanilla MAD reproduction
Phase-1: Injection and Executable Arms	Month 2 (Wk 5–8)	Injection protocol, B3 and B4-matched arms, Proposition 1 proof, month-end pilot
Phase-2: Trust Mechanism Build	Months 3–4 (Wk 9–16)	Claim decomposition, claim-domain retrieval, trust function v1, B8-no-trust and B8 arms
Phase-2 to 3: Mid-Project	Month 5 (Wk 17–20)	Design freeze, dry run on one dataset, FYDP-1 defence preparation, and review feedback fixes
Phase-3a: Main Experiments	Month 6 (Wk 21–24)	Core conditions on primary datasets, two seeds, 95% confidence intervals, and effect size report
Phase-3b: Ablations and Scaling	Month 7 (Wk 25–28)	Four ablations, α/β sweep, $N \in \{2, 3, 5\}$, and results freeze
Phase-4: Human Evaluation and Analysis	Month 8 (Wk 29–32)	Human evaluation with $n = 60$, trust AUROC/Brier calibration, failure analysis, and buffer for slippage
Phase-5: Writing and Defence	Months 9–10 (Wk 33–40)	Thesis drafting, reproducibility package, FYDP-2 defence preparation, paper submission Aug–Dec 2027

Table 3.2: Review gates and their checkpoints.

Gate	Due	Checkpoint
Gate 0	End of Month 1	Vanilla MAD reproduces the Du et al. baseline on a GPQA [24] slice, and the reference list is frozen
Gate 1 (Go/No-Go)	End of Month 2	Injection validation reaches $\kappa \geq 0.75$, baseline CCR sits at 0.30 or higher, pilot executed
Gate 2	End of Month 4	Trust mechanism plus claim-domain retrieval plus the four executable arms, first CCR and MPR on one dataset
FYDP-1 Defence	Month 5	Mid-project report with design freeze and initial results
Gate 3	End of Month 7	Full results freeze with confidence intervals and effect sizes
Thesis and FYDP-2	Months 9–10	Thesis completed, final defence, paper submission Aug–Dec 2027

The reason we have ordered phases sequentially is because there are dependency relationships. Specifically, vanilla MAD reproduction in Phase-0 must finish prior to when injection validation will be performed in Phase-1. Therefore, since the trust function in Phase-2 relies upon an injection harness developed in Phase-1, we cannot begin Phase-2 until after Phase-1 has finished. Similarly, main experiments in Phase-3a must wait for both the completion of the trust function and the completion of the dry run in Phase-2 to 3. Moreover, human evaluation of the results (which occurs in Phase-4) require the frozen results produced in Phase-3b.

However, two buffers act as insurance policies along the critical path of our project. There is first buffer located at Month 5 with respect to a dry run of one dataset being used prior to defending the project; and secondly, there is another buffer located at Month 8 which provides us with an additional month should things slip. These buffers exist at locations along the critical path where a late failure could potentially cause us to miss a hard deadline. As such, weekly meetings and Git version control allow us to track our progress, and each gate review allows us to determine whether to "Go" or "Fix" before we can proceed into the next phase.

Phase-0 spans weeks 1 through week 4. Weeks 1 and 2 are focused on freezing literature and locking reference lists, with each paper being evaluated against their respective DOIs. Week 3 sets up vLLM serving and the Python debate loop. Week 4 evaluates all three models on a small GPQA [24] slice. We need to reproduce Du et al. results on that same slice with our vanilla MAD baseline before moving forward with Phase-1.

Table 3.3: SWOT analysis for the project plan.

Strengths	**Weaknesses**
Clear gap on sycophantic consensus, bounded trust design with no fine-tuning, six-member team with mixed skills, and expert supervision that keeps scope tight. The design needs no new training and runs on a single rented GPU.	Sparse retrieval on rare claims, imperfect claim split, correlated reasoning errors between agents, and a limited GPU budget for the pre-registered 1,280 GPU-hour matrix. Small sample noise in early pilots can also hide real gains.
Opportunities	**Threats**
Growing interest in multi-agent debate and sycophancy, open datasets GPQA and MMLU-Pro, free retrieval APIs from PubMed, arXiv, Semantic Scholar and OpenAlex, and affordable cloud GPU rental with Blackwell FP8 support.	API rate limitations/downtime, price updates, sampling errors due to time slips from complex integrations and evaluation noise in small sample sizes; all contribute to the reliability of the results.

Phase-1 runs Week 5 to Week 8. Week 5 builds the injection harness that inserts a fabricated wrong consensus between rounds. Week 6 to Week 8 validate the injection with kappa checks and run the B3 and B4-matched arms. The month ends with a pilot that must reach CCR 0.30 or higher to pass Gate 1. Phase-2 spans Week 9 to Week 16. Week 9 to Week 12 build claim decomposition. Claim-domain retrieval with cross-encoder reranking is added in Week 13 to Week 16.

Weeks 17–20 (Phase-2 to Phase-3) freeze the design and conduct a "dry" run of a single dataset for the purpose of mid-project review. Weeks 21–24 (Phase-3a), use core conditions on primary datasets with two random seeds and 95% confidence intervals. Weeks 25–28 (Phase-3b), use 4 ablation points and sweep through N and α/β values. Weeks 29–32 (Phase-4), conduct human evaluations with $n = 60$. Finally, Weeks 33–40 (Phase-5) will be devoted to writing the thesis, preparing the final defence, and creating a reproducibility package, with the paper submission planned for August–December 2027.

The pre-registered matrix is budgeted at 1,280 GPU-h under the gate's conservative cap of 400 questions per dataset, or about 1,067 GPU-h under the per-dataset caps of 400 GPQA-Main, 400 MMLU-Pro, and 200 HLE questions. The experiment plan estimates the Core run at USD 420–580 all-in and a Full-run top-up of USD 900–1,200 over the Core envelope. GPU time is concentrated in Phase-3a and Phase-3b. If a retrieval API fails, the run falls back to cached evidence or OpenAlex; a Gate-2 dry-run failure does not move the final defence date because the Month-8 buffer absorbs it.

3.4 Task Allocation

Six members split the work along the modules of the framework. Table 3.4 fixes who owns which primary modules, and Figure 3.5 shows when each member is active across the ten months.

Table 3.4: Module ownership across the six team members.

Member	Primary modules
Md. Atikur Rahaman (Leader)	Architecture, trust mechanism, coordination
Rakibul Hasan	Vanilla MAD reproduction, injection protocol
Md. Salman Rohoman Nayeem	Claim decomposition, claim-domain retrieval
Pratay Paul	Evaluation harness, CCR/MPR, trust AUROC/Brier
Yousuf Kamal Himel	Executable arms B3/B4-matched/B8-no-trust/B8; B1–B10 registry appendix; vLLM serving
Mst. Farjana Akter Limu	Evidence APIs, dashboard, result packages

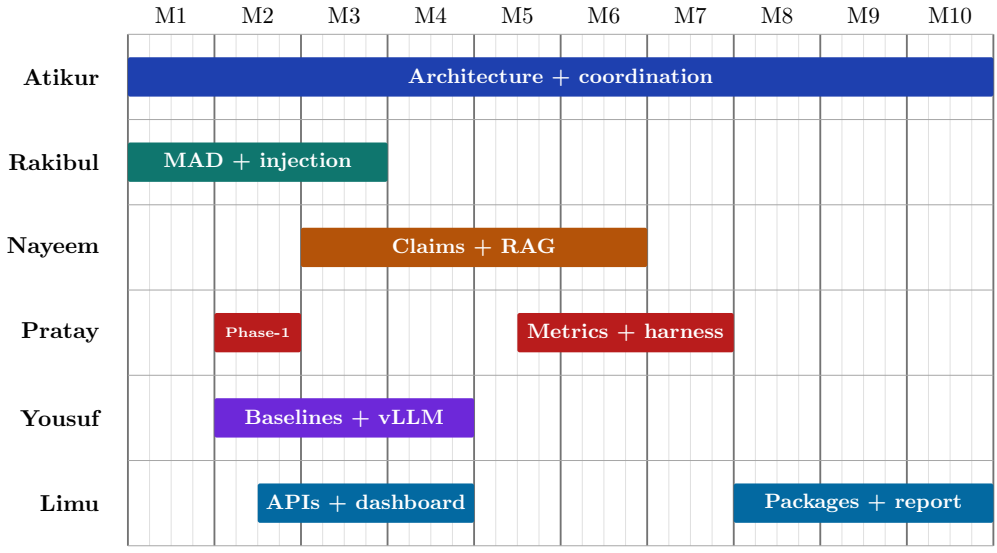


Figure 3.5: Gantt view of member activity across the ten project months, following the phase plan in Table 3.1.

Each task has an identified owner and a simple deliverable. As the owner matches the phase in Table 3.1;all phases will have at least one accountable team member. All

dependencies are ordered. Vanilla reproduction will be completed before Injection begins; the Injection Harness completes before Trust Build begins. Dry Run must be passed before Main Experiments begin; Human Evaluation waits for Results Freeze. Writing utilizes the frozen results and occurs last. Each Module is reviewed by the Leader prior to being marked complete. A weekly Git commit and brief stand-up meeting monitor slippage; if any gates are missed, a "fix" week will occur prior to beginning the next phase.

3.5 Summary

Chapter 3 fixed the functional and nonfunctional requirements and drew the system boundary in the context diagram. The data flow diagram and the interface design follow. The design section compared the main alternatives and recorded why each choice was made. The project plan then locks eight phases and six gates that carry the work to the final defence. Implementation and evaluation come next.

Chapter 4

Implementation and Validation

This chapter records the implementation of the trust-calibrated deliberation framework and the validation evidence available at the mid-project review. It separates three kinds of evidence. The first is what ran on real hardware: Gate 0, a vanilla multi-agent debate on ten GPQA questions. The second is what the P0 gate checks without a GPU: option rendering, answer parsing, trust bounds, endpoints, dataset pins, claim routing, the manifest guard, and the matrix budget. The third is the test suite, which exercises the same modules on synthetic and mock inputs. None of these is an efficacy result. The comparative matrix has not been run, so the chapter makes no claim that trust weighting reduces collapse or improves accuracy.

4.1 Execution Environment

The pipeline runs in two configurations. The development configuration is the one used for Gate 0; the final configuration is pinned but has not been smoke-tested. Table 4.1 lists both. The development card is an NVIDIA RTX A6000 with 48GB of memory (about 45GB usable), hosted on a Vast.ai instance with 12 vCPU, 72GB of system RAM, and a 100GB container disk. The operating system was Ubuntu 22.04, Python was 3.12.3, PyTorch was 2.11.0 with CUDA 12.8, and vLLM was the nightly build `0.29.1rc1.dev128`. The final card is an NVIDIA RTX PRO 6000 Blackwell with 96GB. The pinned checkpoints are Qwen3.6-27B-FP8 (26.0GB), Gemma 4 26B-A4B-FP8-dynamic (28.67GB), and Mistral Small 3.2 24B-AWQ (15.06GB), totalling 69.73GB. The final context length is 8192 tokens, and the three servers are configured to use 0.90 of the card in total.

The development stack uses 4-bit checkpoints because the A6000 is an Ampere card. The official Ministral FP8 checkpoint cannot run on Ampere: vLLM's W8A8 sm80 kernel fails. The development agent therefore uses an AWQ 4-bit checkpoint of the same model. This is a precision change, not a model change, and it is recorded in the model configuration. The final stack uses two FP8 checkpoints and one AWQ checkpoint. The total weight footprint of 69.73GB leaves room for the KV cache on the 96GB card, but the final stack has not been exercised on hardware; the smoke test remains outstanding.

Table 4.1: Development and final serving configurations. The final stack is pinned but was not smoke-tested at the time of this report.

Item	Development (Gate 0)	Final (pinned)
GPU	RTX A6000 48GB (45GB usable)	RTX PRO 6000 Blackwell 96GB
Host	12 vCPU, 72GB RAM, 100GB disk	Not yet provisioned
Software	Ubuntu 22.04, Python 3.12.3, PyTorch 2.11.0, CUDA 12.8	Python 3.10 or newer; image smoke test pending
vLLM	0.29.1rc1.dev128	Pinned with the final image; smoke test pending
Agent 1	Qwen3.5-9B-AWQ, 4-bit AWQ	Qwen3.6-27B-FP8, 26.0GB
Agent 2	Gemma 4 12B, QAT w4a16	Gemma 4 26B-A4B, FP8 dynamic, 28.67GB
Agent 3	Ministral-3-14B, AWQ 4-bit	Mistral Small 3.2 24B, AWQ, 15.06GB
Context	4096 tokens	8192 tokens
Serving	OpenAI-compatible, ports 8000–8002, sequential readiness-gated startup	Same interface; final smoke test pending

All model calls go through the OpenAI-compatible chat-completion interface. The client retries failed calls up to three attempts with exponential backoff, does not retry an HTTP 400, and treats an empty completion as a failed attempt. Generation uses temperature 0.7 and a 1024-token output cap in the development configuration. Peer positions are truncated to 1500 characters in revision prompts so that the three-agent transcript fits the context window. The first Gate 0 run exposed this constraint: an early prompt overflow combined 2049 input tokens with a 2048-token output budget against a 4096-token context. The cap and the peer truncation are the fix, and both are now part of the configuration rather than an operator habit.

4.2 Pipeline Implementation

4.2.1 Debate Loop

The debate loop is a plain Python state machine rather than a graph framework. It takes a question and an optional list of answer choices, creates one client per agent, and runs $K = 3$ rounds. Round one collects independent positions. In rounds two and three, each agent sees the question, the rendered options, its own previous position, its own current trust weight, and the other agents' positions truncated to 1500 characters. The agent does not see the other agents' trust weights. The loop stores the transcript, the parsed answers

for each round, the injection record, the per-claim verdicts, the evidence passages, and the final aggregation.

The number of agents is three, and the three model families differ. The design assumes that different pre-training pipelines reduce correlated errors, but this assumption is not tested in this report. The loop also supports two and five agents for the planned N-scaling ablation; the trust operator is tested for $N = 1, 3, 5$.

4.2.2 Options and Answer Parsing

The GPQA loader places the correct answer first, and MMLU-Pro keeps the dataset order. Neither order may reach a prompt. The option renderer shuffles the choices with a per-question seed derived from a SHA-256 hash of the question identifier and the run seed, so the order is reproducible across processes and resume does not change it. The record stores the raw options, the rendered options, the permutation, and the rendered letter of the correct option. Free-form items, such as HLE, receive the literal instruction "Answer format: free text" and no fabricated choices.

The parser prefers an explicit answer span: an `Answer:` or `Final answer:` line, a boxed expression, or the last line containing a letter or number. It handles caret and LaTeX scientific notation, strips units before numeric comparison, and maps a letter or option text to the rendered option. A 50-case corpus with gold parses is part of the gate; the gate reports 50/50 exact matches on that corpus and the probes `10^-4 eV`, `7.3 x 10^-7`, `The answer is B`, and `Product 3 contains 11`. Every record carries a `parser_version` so a parser change is auditable. This matters because Gate 0 showed that 17 of 90 generations opened with a formatting-only line such as `**Answer:**` and put the real answer on the next line.

4.2.3 Claim Decomposition

Agents are prompted to tag claims as `<claim id="cX">...</claim>` in the same generation. The parser extracts the tagged spans; if no tags are present, it falls back to sentence splitting, keeps items of at least three words, and caps the fallback at 20 claims. The trust configuration caps the number of claims used per agent at eight. The fallback is load-bearing. In Gate 0, Qwen and Gemma tagged every generation (30/30 each), while Ministral left 5 of 30 generations untagged; without the fallback, those positions would contribute no verifiable claims.

The claim text is what the retrieval and verifier layers see. Untagged prose is not verified. A judgment-style claim usually returns unverifiable and therefore does not move trust. This is a deliberate limit: the mechanism only acts on claims that the evidence layer can check.

4.2.4 Claim-Domain Retrieval

Retrieval is routed by claim domain rather than by agent identity. Biomedical and clinical claims query PubMed and Semantic Scholar; physics, chemistry, mathematics, and computer-science claims query arXiv and Semantic Scholar; other claims query Semantic Scholar or OpenAlex. This replaces the earlier design in which agent 1 always used PubMed, agent 2 always used arXiv, and agent 3 always used Semantic Scholar. The earlier design confounded source, domain, and model family: an agent could appear better because its assigned source covered the question's domain more easily. The current routing gives all agents the same evidence for the same claim, and the source mix comes from the claim set rather than from the agent's identity.

Retrieved passages are deduplicated by DOI, arXiv identifier, or title. The retriever takes a small candidate set and selects the passages shown to the agent; the trust configuration uses three passages per claim. The live PubMed, arXiv, Semantic Scholar, and OpenAlex calls are not exercised by the test suite. Retrieval failures are recorded rather than fatal, and the claim becomes unverifiable when no passage is returned. The retrieval layer is therefore implemented and routed, but its live coverage and rate-limit behaviour remain an operational risk.

4.2.5 Verifier Interface

The verifier is a separate interface with four implementations behind it: lexical overlap, cross-encoder, natural-language inference, and an LLM judge. The default is lexical. A supported verdict requires support in the retrieved passage; a contradicted verdict requires an explicit contradiction; no evidence yields unverifiable. Contested evidence is reported separately and is not counted as support or contradiction. The interface records the verifier name and version with each verdict.

The lexical default is a diagnostic, not a validated scientific verifier. It scores claim-term coverage and can misread numeric or negated statements. The cross-encoder is optional and requires `sentence-transformers`; the LLM-judge hook exists but is not passed by the debate runner in the current branch. The human audit of 150–200 claim-passage pairs has not been completed. For that reason, this report makes no claim about verifier accuracy, and the planned matrix treats verifier quality as a moderator rather than a solved component.

4.2.6 Trust Update

The trust update follows the design in Chapter 3. Let $w_i^{(t)}$ be agent i's weight at round t, V_i the number of its claims that the verifier supports, H_i the number it contradicts, and n_i the number checked. The operator accumulates evidence in the log domain,

$$\ell_i^{(t+1)} = \log w_i^{(t)} + \alpha \frac{V_i}{n_i} - \beta \frac{H_i}{n_i}, \tag{4.1}$$

applies one softmax, and projects the result onto the polytope

$$\mathcal{W} = \left\{ w \in \mathbb{R}^N : w_i \geq f,\ \sum_{i=1}^{N} w_i = 1,\ w_i \leq \max(c, 1/N) \right\}. \tag{4.2}$$

The defaults are $\alpha = 0.5$, $\beta = 0.3$, $f = 0.1$, and $c = 0.9$. The projection guarantees $w_i \geq f$ and $\sum_i w_i = 1$ for every finite input. The ceiling is relaxed to $1/N$ only when $Nc < 1$, a case with no feasible vector; $N = 1$ returns [1.0] because that is the only feasible vector. Neutral evidence, $V_i = H_i = 0$, is an identity for an already feasible vector: the weights do not drift toward uniform. Dividing by n_i cancels verbosity, so duplicating every claim does not increase an agent's weight.

The operator is implemented in `trust/update.py`. The P0 gate runs 30,000 random draws for $N = 1, 3, 5$ and checks the floor and the sum. It also checks neutral evidence and verbosity invariance. The debate loop passes the per-agent claim counts to the operator, calls it after each evidence round, and performs a closing pass on the final positions, so the vector used by aggregation is computed from the final positions rather than from an earlier round. The result carries `trust_for_positions` and `trust_status`, and the runner copies those fields into every record.

4.2.7 Weighted Aggregation

Aggregation is a weighted plurality, not a majority vote. Each agent's final parsed position is mapped to a canonical answer group. For each group a, the score is

$$S(a) = \sum_{i:\, p_i = a} w_i, \tag{4.3}$$

and the group with the largest score wins. Ties resolve to the first observed answer. The majority answer is computed and stored alongside the weighted answer so that the same transcript can be scored under both rules; it does not choose the treatment's final answer. This is what lets the pilot compare the weighted and unweighted winners without re-running the models.

4.2.8 Injection Protocol

The injection path is a one-shot stress test. After round one, the builder checks whether the panel is divergent, whether every answer is checkable as a multiple-choice letter or a number, and whether a target exists. A unanimous panel is ineligible, as is a panel with fewer than two agents. The eligible plan fabricates a consensus and targets the agents that disagree with it by default; the `all` scope is also supported. The injected text is appended to the targeted agents' system prompt before round two. Round three carries no injection. Ineligible items are logged with a reason and are not fatal. The injection path is implemented, mock-tested, and wired to the B8-plus-injection arm; it has not been exercised on a real model panel.

4.2.9 Runner, Records, and Manifest

The runner writes one JSON line per completed debate and appends it immediately, so a crash costs at most one question of GPU time. Re-running the same configuration skips completed question identifiers. Every record carries the question identifier, arm, seed, question, gold answer, raw and rendered options, permutation, rendered gold letter, raw answer, parsed answer, parser version, per-round answers, transcript, injection record, trust trajectory, final trust, verdicts, evidence, aggregation, and timing. The schema validator requires the final answer, its source, the parser version, and a valid trust simplex; it rejects a record rather than writing a plausible but unobserved value.

The manifest layer hashes the configuration files, the run configuration, the model checkpoints, the prompt file, the pinned dataset revisions, the code commit, the temperature, and the seed. The hash excludes the creation time, so re-running the same configuration later resumes instead of forking. The resume guard refuses to append when a result-affecting field differs, and it names the differing keys. The P0 gate exercises the guard with an identical manifest and a changed manifest; the changed manifest is refused. The runner stamps every record with the manifest hash, so the contract is implemented, gate-tested, and present in the records written by the runner.

4.3 P0 Hardening Changes

Table 4.2 lists the P0 changes that are present in the P0 release. The table uses "gate-tested" for a path that the P0 gate exercises without a GPU. The gate reports 13 passed, 0 failed, and 0 pending checks. The verifier human audit, the EG-SSS calibration set, the paid Core run, and the final-stack smoke test are outside the gate and remain outstanding.

Three changes need a word of explanation because they alter what the earlier draft claimed. First, the trust operator is no longer a clamp followed by renormalization. The old order could leave a weight below the floor after renormalization; the projection makes the floor a property of the output rather than a hope. Second, the option order is no longer the dataset order. The correct answer was always option A in the GPQA loader, and the prompts did not contain the options at all. The renderer and the scorer now use the same permutation. Third, the matrix is no longer the 15-condition grid. The executable matrix is four arms $\times$ three datasets $\times$ two seeds, or 24 runs, at 1,280 GPU-h under the gate's conservative cap. The six-arm target adds B8-placebo and B8-oracle through WP-18; the pre-registration names both and states that the thesis reports the six-arm design as incomplete rather than silently dropping the contrasts if they are not added before unblinding.

4.4 Testing and Validation

4.4.1 Test Suite

The branch records 514 passing tests. The suite covers the parser and answer matcher, the option renderer, the inference client's retry and failure behaviour, the debate loop and injection targeting, the trust bounds, the endpoint definitions, the statistics helpers, the dataset loaders, the manifest and resume guard, and the mock end-to-end path. The tests are colocated with the modules they exercise. A passing suite means the contracts hold on the tested inputs; it does not mean the framework has been validated on a representative question set.

The P0 gate is a separate 13-check script that runs before any paid run. It checks the test suite, the record schema, options in prompts, the parser corpus, trust bounds, neutral and verbosity behaviour, the B8 versus B8-no-trust prompt diff, the B8-plus-injection mock, the endpoint summary, dataset pins, claim routing, the manifest guard, and the matrix budget. It reports 13 passed, 0 failed, and 0 pending. The gate is an engineering readiness check, not a result.

4.4.2 Mock End-to-End Path

The repository includes a mock vLLM server with three modes: agree, divergent, and untagged. The integration test starts the mock panel, runs an arm against it, and asserts the designed injection-collapse scenario. This path is how the injection, trust, aggregation, and record code are exercised without a GPU. It is also the path used to demonstrate the pipeline when a GPU is unavailable. The mock server does not test model quality, retrieval quality, or verifier accuracy; it tests orchestration and data flow.

4.4.3 Gate 0

Gate 0 is the only path that has run on real GPUs. It ran vanilla multi-agent debate on ten GPQA-Diamond questions with three agents and three rounds. Table 4.3 lists the measurements. All ten debates completed three rounds, all 90 positions were non-empty, and the mean debate took 370.8 seconds with a standard deviation of 21.5 seconds. The fastest debate took 334.8 seconds and the slowest 404.4 seconds. The run generated 275,668 characters, used about 1.0 GPU-hour, and cost approximately USD 4.29 including model and image ingress.

Gate 0 proves that the serving layer, the three-agent loop, the revision rounds, the parser, and the result logging work end to end at 4-bit precision on one GPU. It does not prove that the trust mechanism changes any decision. The trust trajectory is null because the run was vanilla MAD. The 5 of 30 untagged Ministral generations show that the fallback parser is necessary, not decorative. The 21.5-second standard deviation shows that per-question cost is predictable enough for planning, not that the matrix will finish in a fixed time.

4.5 Limitations and Reproducibility

The implementation has five limitations that matter for the next phase. First, the verifier is lexical by default. The cross-encoder is optional, the NLI and LLM-judge implementations are interfaces rather than validated components, and the 150–200-pair human audit has not been run. A trust signal built on a weak verifier can amplify noise; the pre-registration treats verifier quality as a moderator for exactly this reason. Second, the live retrieval APIs have not been exercised by the test suite. The routing and deduplication logic are tested, but rate limits, malformed responses, and source coverage are operational unknowns. Third, the final stack is pinned but not smoke-tested. The 69.73GB weight total is a configuration check, not a serving result. Fourth, the runner-side record adapter is wired: the debate loop performs a closing pass on the final positions and exposes `trust_for_positions`, and the runner copies that vector, the final-answer fields, and the manifest hash into every record. Fifth, the only real-model run is Gate 0. No paid run has exercised the trust path, the verifier path, the injection path, or the matrix.

The reproducibility package is designed around five artifacts. The code is released under the MIT licence (see `LICENSE` at the repository root). The model configuration pins every checkpoint and revision, and the dataset configuration pins every revision as a 40-character commit hash. The seed is recorded in the run configuration and in the manifest, and the per-question option permutation is derived from the seed and the question identifier. The manifest records the configuration hash, the model checkpoints, the prompt hash, the dataset revisions, the code commit, the temperature, and the seed, and the resume guard refuses a mismatch. The pre-registration fixes the four executable arms, the three datasets, the two seeds, the primary and co-primary endpoints, the analysis plan, and the stopping rules before the matrix runs.

Gated datasets are shared as identifiers and hashes only. GPQA and HLE require a click-through agreement, and their item text is not redistributed. MMLU-Pro is public. The code release does not include model weights or raw dataset rows. The result package includes the transcripts and the evidence trail, which is what a reader needs to audit a decision without receiving the benchmark items.

4.6 Summary

The implementation is complete enough to run the debate loop, render and score multiple-choice options, decompose claims, route claims to domain-specific sources, update a bounded trust vector, aggregate a weighted answer, log a record, and validate the result schema. Gate 0 validates the base loop on ten GPQA questions. The P0 gate validates the wiring that Gate 0 did not cover, and the 514-test suite covers the contracts on synthetic and mock inputs. The remaining work is the verifier human audit, the EG-SSS calibration set, the WP-18 placebo/oracle slice, the final-stack smoke test, and the paid matrix. Those are the conditions for an efficacy claim, and none of them is met in this report.

Table 4.2: P0 changes present in the implementation branch. Gate-tested means the path is exercised by the 13-check P0 gate, not that it has been validated on a paid run.

P0	Change	Module or evidence	Status
0	Answer options in every multiple-choice prompt; seeded per-question permutation; free-text fallback	Option renderer and prompts	Gate-tested: 5 questions $\times$ 20 seeds; every option exactly once; gold letters vary
1	Answer-span parser and scorer; 50-case corpus; parser version	Parser and scorer	Gate-tested: 50/50 exact; notation and unit probes pass
2	Final-answer fields and accuracy endpoint	Record schema, endpoints	Endpoint gate-tested on fixtures; runner stamps final-answer fields into every record
4	Eight endpoints with raw counts and confidence intervals	Endpoints module	Gate-tested: 8 endpoints on 6 records
5	Logit accumulation, one softmax, simplex projection, claim-count normalization; closing pass on final positions	Trust update and debate loop	Gate-tested: 30,000 draws for $N = 1, 3, 5$; neutral and verbosity checks pass; WP-03b timing landed
6	Verifier interface with lexical, cross-encoder, NLI, and LLM-judge implementations	Verifier interface	Interface gate-tested; lexical default; human audit pending
8	Pinned dataset revisions and claim-domain routing	Dataset config and routing	Gate-tested: 8 datasets pinned; bio routes to PubMed, math does not
10	Matrix cut to four executable arms; preregistration	Matrix config and preregistration	Gate-tested: 24 rows, 1,280 GPU-h, version 1.0; six-arm target is WP-18
11	Manifest and resume guard	Manifest module	Gate-tested: mismatch refused, identical manifest accepted; runner stamps the manifest hash

Table 4.3: Gate 0 measurements for the vanilla debate loop. These values validate serving, orchestration, and parsing; they do not measure trust or collapse.

Check	Result
Debates completed	10/10
Rounds completed	3/3 on all ten debates
Non-empty positions	90/90
Generated text	275,668 characters (about 69K tokens)
Total runtime	3,708.1 seconds (61.8 minutes)
Mean per debate	370.8 seconds
Standard deviation	21.5 seconds (5.8%)
Fastest and slowest debate	334.8 seconds and 404.4 seconds
Claim-tag compliance	Qwen 30/30, Gemma 30/30, Ministral 25/30
Trust trajectory	Null; Gate 0 ran before trust wiring
Invoice	Approximately USD 4.29, including ingress

Chapter 5

Standards and Design Constraints

This chapter documents the standards that the framework adheres to, The constraints that have shaped its design and budgetary costs of the Experiments; and concludes by mapping the project against the Criteria for complex engineering problems and providing a brief summary.

5.1 Compliance with the Standards

Standards play out in three areas - the software stack, the hardware the Framework executes upon, and the communication among components. There were Alternative options available in each area and these alternatives are explained Below.

5.1.1 Software Standards

Python was chosen because the LLM serving and evaluation ecosystem is concentrated in Python: Hugging Face libraries, vLLM, PyTorch, and the evaluation tools all expose Python interfaces. Java and C++ are faster, but neither has the same pre-existing support for loading and serving current LLM checkpoints. The framework is a research prototype rather than a real-time production service, so ecosystem fit matters more than raw speed. The Gate 0 environment used Python 3.12.3; the repository requires Python 3.10 or newer. Surveys of LLM-based multi-agent systems describe this design space in terms of workflow, infrastructure, and open challenges [14]; the decisions below follow that structure.

vLLM is built on PyTorch and directly supports the Qwen, Gemma, and Mistral families. TensorFlow and JAX were evaluated as alternatives; TensorFlow's LLM serving story is weaker, and JAX would require rewriting the pipeline without a clear advantage.

vLLM serves models through an OpenAI-compatible API. The orchestration code can therefore call closed APIs and locally served models without modification. PagedAttention manages KV-cache memory for concurrent requests, and the Blackwell FP8 path supplies the throughput needed for the three-agent load. Ollama is simpler to set up but slower under concurrent requests; llama.cpp runs well on CPU but does not match the GPU

throughput; TGI is a close alternative, but vLLM fit the existing ecosystem and exposed the OpenAI-compatible surface.

Cross-encoders rerank passages returned by retrieval. The implementation uses a sentence-transformers cross-encoder when the optional dependency is installed; the default verdict path is lexical and does not require it. BM25 is an alternative ranker, but it scores lexical overlap rather than semantic relevance. Since the verifier must judge whether a passage supports a claim, the cross-encoder is an optional reranker, not the verdict model. The web dashboard is planned with FastAPI, and result packages are JSON objects following RFC 8259 that are validated against a schema. Codebase changes are tracked with git, and LaTeX (Overleaf) generates this document.

5.1.2 Hardware Standards

Experiments are conducted on a single NVIDIA RTX PRO 6000 Blackwell GPU with 96 GB GDDR7 memory. Memory capacity and architecture were the key considerations. The development models (Qwen3.5-9B, Gemma 4 12B, Ministral-3-14B) run as 4-bit checkpoints on the 48GB development card. The final models (Qwen3.6-27B, Gemma 4 26B, Mistral Small 3.2 24B) total about 77 billion parameters. The pinned final checkpoints are two FP8 models and one AWQ model, totalling 69.73GB, so they fit on one 96GB card with room for the KV cache.

An initial alternative was the RTX A6000 80GB. It has enough memory for the development stack at 4-bit precision and rents at about $0.263 per hour, but it is an Ampere card: it cannot run the FP8 final checkpoints and would require a fully 4-bit final stack. Faster still is the H100 80GB; however it costs roughly double as much and offers none of what our framework requires. Finally, the L40S at 48 GB and the consumer RTX 5090 at 32 GB do not have enough memory to hold the full final model trio. Thus the RTX PRO 6000 provides enough memory for the pinned stack, supports FP8, and can be rented within the student budget. To stage models prior to moving them to the GPU the host computer needs at least 64 GB of system RAM and NVMe disk storage.

5.1.3 Communication Standards

All remote communication between the framework and its components uses HTTPS. Local vLLM servers expose HTTP on the host. The model server presents an OpenAI-compatible chat-completion API endpoint that is REST-based with JSON payloads. gRPC was another option for model serving; it has lower latencies in benchmark tests, but an OpenAI-compatible surface lets the framework use closed APIs and local vLLM-served models without changing the orchestration code.

Each source of evidence is accessed through its public REST interface: PubMed E-utilities, the arXiv API, the Semantic Scholar Graph API, and OpenAlex as a fallback. Each returns JSON or XML over HTTPS; the framework handles rate limits with retries capped according to the requirements. Web sockets are not used in the P0 implementation;

live dashboard updates are planned.

5.2 Design Constraints

5.2.1 Economic Constraint

The matrix is budgeted at 1,280 GPU-h under the gate's conservative cap, or about 1,067 GPU-h under the per-dataset caps. The experiment plan estimates the Core run at USD 420–580 all-in and a Full-run top-up of USD 900–1,200 over the Core envelope. Retrieval APIs are free at their public tiers, GPQA and MMLU-Pro are public, and the code is inference-only, so no training compute is required. The main economic risk is a GPU price increase or a retrieval API moving to a paid tier; both are covered by the contingency in the budget.

5.2.2 Environmental Constraint

The project avoids training and uses quantized inference, which keeps the compute footprint to the rented GPU hours above. Energy use was not metered during Gate 0, so the report makes no quantitative energy claim. The main environmental cost is GPU electricity and the storage of model weights and logs. The run plan reuses the model cache and copies result packages off the host before the instance is destroyed.

5.2.3 Ethical Constraint

The framework generates answers to scientific questions and can be wrong. It is a research prototype, not a source of clinical, legal, or safety-critical advice. The injection experiment fabricates a consensus, but only inside a controlled benchmark; no human subjects are involved. Dataset licences are recorded, gated datasets are not redistributed, and the result package keeps the evidence trail so a reader can inspect why an answer was selected.

5.2.4 Health and Safety Constraint

No physical system is controlled. The relevant safety risk is that a user treats a generated scientific answer as authoritative. The design mitigates this by returning the evidence trail and by evaluating abstention on retrieval-resistant items. It does not eliminate the risk, and the report does not claim deployment readiness.

5.2.5 Social Constraint

The work targets a social problem: confident but unsupported answers can mislead users. The framework is intended to make disagreement and evidence visible rather than to hide it behind a single vote. Access is a constraint: the code and configuration are intended for open release, while gated datasets are shared as identifiers and hashes only.

5.2.6 Political Constraint

The project has no direct political constituency. Scientific misinformation is a public issue, and the framework could be used to support or contest claims. The evaluation therefore reports failure cases and does not claim that the mechanism is neutral.

5.2.7 Sustainability

The main sustainability choice is inference-only operation on quantized checkpoints, which avoids training runs. The matrix is bounded by the pre-registration, and the runner resumes instead of repeating completed questions. Longer-term use would need a cost and energy audit; that audit is not part of this report.

5.3 Cost Analysis

The budget covers GPU rental, storage, and the free tiers of the retrieval APIs; no training compute is purchased. The pre-registered matrix is four arms × three datasets × two seeds, or 24 runs. At the gate's conservative 400 questions per run and 8 minutes per debate, the matrix is 1,280 GPU-h; the per-dataset caps of 400 GPQA-Main, 400 MMLU-Pro, and 200 HLE questions give about 1,067 GPU-h. The experiment plan estimates the Core run, including pilot and sweep, at USD 420–580 all-in, and a Full-run top-up of USD 900–1,200 over the Core envelope. The Full run is the six-arm target: 36 runs at 300 questions per dataset, or 10,800 debates, under a 1,440 GPU-h envelope if B8-oracle is defined as gold-answer trust; the count is recomputed if WP-18 defines it as gold-evidence-only on EG-SSS. The Core run is a prefix of the Full run only if the model stack matches. The earlier 300-hour figure described the pre-P0 plan and is superseded. Table 5.1 gives the planning envelope; Table 5.2 separates the executable matrix from the larger scenarios.

Storage and bandwidth are included in the scenario envelopes rather than billed separately. The retrieval APIs are free at their public tiers, but rate limits can increase wall-clock time. GPQA [24] and MMLU-Pro [25] are public; HLE and GPQA are gated and are not redistributed. Printing (approximately $15) and internet connectivity are treated as fixed and excluded from the compute envelope. Overleaf is provided to the team, so it adds no cost.

An alternative budget using the RTX A6000 80GB was also considered. At about $0.263 per hour, the same 1,280 GPU-h matrix would cost about $337 in GPU time, before storage and overhead. The A6000 is cheaper but has no FP8 path, so the final stack would have to be fully 4-bit and the final-stack replication would not exercise the

Table 5.1: Planning envelope for the project.

Item	Basis	Cost (USD)
Executable matrix GPU time	1,067–1,280 GPU-h	within Core-run envelope
Core run (matrix, pilot, sweep)	1,160 GPU-h planning	420–580 all-in
Full run (six arms, 300 per dataset)	36 runs; 1,440 GPU-h envelope	900–1,200 top-up
Retrieval APIs (PubMed, arXiv, S2, OpenAlex)	public tiers	0
Datasets (GPQA, MMLU-Pro, HLE)	public or gated	0
Storage and bandwidth	included in the envelopes	included
Planning range		**420–1,780**

pinned FP8 checkpoints. The RTX PRO 6000 was chosen because the Blackwell FP8 path keeps the pipeline simple and avoids tighter quantization or sequential model loading.

There is no revenue stream anticipated for this project. The project is an academic deliverable produced by a university team with funding provided through grants awarded to the university and team. The expected products are a written report and an open-source framework. The alternate A100 budget is shown in Table 5.3.

A ten percent buffer on the Core-run envelope is about USD 42–58, and on the Full-run top-up about USD 90–120. The buffer covers price increases and extended operating time if a run fails or a gate needs a repeat before the end of the plan period.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

The project is assessed against the seven attributes defined in the Washington Accord criteria. Six are covered; P6 is not covered because the project has no external stakeholder. Each attribute is mapped to subsections in Table 5.4, which follows.

Table 5.2: Compute plan and the one completed GPU run.

Work item	GPU hours	Evidence
Gate 0 (completed)	≈ 1.0	3,708.1 s total; invoice USD 4.29 including ingress
Executable P0 matrix	1,067–1,280	24 runs; gate cap or per-dataset caps
Core run (pilot and sweep included)	1,160 planning	experiment-plan Core-run scenario
Full run	1,440 (36 runs)	USD 900–1,200 top-up over Core

Table 5.3: Alternate Core/Full budget on the RTX A6000 80GB (4-bit Dev stack).

Item	Quantity	Unit Cost (USD)	Total (USD)
RTX A6000 80GB rental	1,280 hours	0.263 / hour	337
Retrieval APIs	—	0 (free tier)	0
Datasets	—	0 (public)	0
Storage	200 GB	0.20 / GB-month	40
Total			**377**

P1: Depth of Knowledge Required

This project required knowledge of multiple areas including machine learning, natural language processing, retrieval-augmented generation, and distributed model serving. Therefore, solving the project required specialized knowledge in these areas, such as the trust update formula through the vLLM serving layer. The coverage for P1 is based on the Washington Accord K1-K8, as detailed in Table 5.5 after P7.

P2: Range of Conflicting Requirements

P2 is covered. The project needed to meet multiple competing requirements. Accuracy competed with latency and cost. More rounds improved the signal but increased both the inference time and the GPU bill. Strong weighting would limit diversity due to trust calibration. The need to cover evidence competed with API rate limits. Therefore, the team needed to resolve these together. Round caps and simplex bounds were established, and verdicts are conservatively resolved. Thus, none of the competing needs dominate.

Table 5.4: Mapping with complex problem solving (Washington Accord P1–P7).

P1 Depth of Knowledge	P2 Range of Conflicting Require-ments	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stakeholder Involvement	P7 Inter-dependence
√	√	√	√	√	×	√

P3: Depth of Analysis

P3 is covered. There is no textbook solution to the problem. The design defines comparative experiments against majority voting, MoA, iMAD, and DebUnc. It also defines metrics for the failure modes: final-answer accuracy, correct-minority retention, collapse under pressure, appropriate revision, and trust calibration. The injection point adds a controlled wrong-consensus condition. The matrix is pre-registered but has not been run, so the depth of the design is what supports P3 at mid-project, not completed comparative results.

P4: Familiarity of Issues

P4 is covered. Multi-Agent LLM Debate is a recent research area. The most similar research appeared between 2024 and 2026, and we are not aware of a standard method for resolving sycophantic collapse in multi-agent LLM debates. At the beginning of the project, the issues were unfamiliar to the team. A review of current literature was conducted before designing the framework, as described in Chapter 2. That review is what supports the P4 assessment.

P5: Extent of Applicable Codes

P5 is covered. General software standards are applicable. JSON (RFC 8259) and HTTP over TLS are applicable. Additionally, since the framework uses an open AI compatible API, the general API standard applies. However, as previously stated, there is currently no applicable code that addresses evidence-grounded trust in LLM debates. As such, this area is only partially standardized and the framework defines its own rules for the trust score and four possible verdicts. Therefore, this combination of existing codes and new conventions make P5 "covered."

P6: Extent of Stakeholder Involvement

P6 is not covered. The project does not have any external stakeholders. The only individuals involved are the supervisor and the examiners. There is no industry partner and no community group that identifies stakeholder needs or evaluates results as part of the development process. While users who will use future scientific QA tools may benefit from this project, they do not participate in either the design or testing phases of the project. The framework was designed and tested as an Academic FYDP (Final Year

Design Project); therefore, the engagement is very narrow and only within the University context. For this reason P6 is considered "not covered", while the other P points are still considered covered.

P7: Interdependence

P7 is covered. The tight interdependencies between the Confidence Gate, Orchestrator, Claim Decomposer, Retrieval Subsystem, Trust Updater and Aggregator make it a candidate for P7. This means that if there is an error in the Retrieval Subsystem, this will be reflected in both Verdicts and ultimately the Final Answer through the Trust Score. Because of these interdependencies, P7 has been marked as "covered" with the addition of the Failure-Isolation Rule identified in Section 3.1.1.

Table 5.5 details the profile. The paragraphs below explain each K for this project and why it is covered or not.

Table 5.5: Knowledge profile mapping for P1 (K1–K8).

Knowledge	Applied	Where used
K1: Natural sciences	√	Scientific questions; evidence from PubMed, arXiv, Semantic Scholar, OpenAlex
K2: Mathematics	√	Logit accumulation, softmax, simplex projection, probability
K3: Engineering fundamentals	√	System design, state machines, failure handling
K4: Specialist knowledge	√	LLMs, multi-agent debate, RAG, sycophancy
K5: Engineering methods	√	Experimental methodology, baseline comparison
K6: Computational methods	√	Python, PyTorch, vLLM, FastAPI
K7: Codes and practices	√	JSON, HTTP, TLS, version control
K8: Research and context	√	Ethical constraints and use in scientific QA

K1: Natural sciences K1 has been addressed. The framework is designed to address scientific inquiries and verify each claim by comparing it to articles on PubMed, arXiv, Semantic Scholar, and OpenAlex. Natural science questions are included in evaluation data sets. The project itself is not an outcome of a new natural science theory. The

problem domain and the evidence base are still sourced from the natural science fields. This is the reason for marking K1 as covered.

K2: Mathematics K2 has been addressed. The trust operator uses logit accumulation, one softmax, and a simplex projection with a floor. Probability and statistical measures are used in the evaluation, including bootstrap intervals and paired tests. Both the operator and the evaluation use the same probability framework.

K3: Engineering fundamentals K3 is covered. State machines (the system's design and its failure handling) are at the heart of what we've done here. Basic engineering design is used by requirements and the confidence gate along with the orchestration logic.

K4: Specialist knowledge K4 is covered. The main areas are large language models, multi-agent debate, retrieval-augmented generation (RAG), and sycophancy. These areas drive how claims are divided, how evidence is retrieved, and how trust is computed.

K5: Engineering methods K5 is covered. An experimental methodology is used with controlled variables. The analysis plan specifies confidence intervals and baseline comparisons. The comparative runs are not yet complete; the evidence at mid-project is the implementation, the test suite, and Gate 0.

K6: Computational methods K6 is covered. Our main application tools are Python, PyTorch, vLLM (to serve the model), and FastAPI (to serve the dashboard). Sentence-transformers (to re-rank passages) and Git (for version control) complete the suite of applications.

K7: Codes and practices K7 has been addressed. JSON under RFC 8259, HTTP with TLS, the OpenAI compatible API, and JSON Schema (as well as many others) were all applied as codes. Standard practices were followed regarding version control and coding style.

K8: Research and context K8 has been addressed. Scientific QA (scientific questions answering) provides us with our ethical bounds on the output from AIs — literature reviews, gap analyses provide the context for our research direction.

The profile is broad rather than deep in one spot. The report applies K4 and K5 in every chapter, and K6 and K7 in the implementation, which shows the range the problem demands.

5.4.2 Engineering Activities

Table 5.6 maps the project to the five engineering activities A1–A5, with rationales below.

Table 5.6: Mapping with complex engineering activities (A1–A5).

A1 Range of resources	A2 Level of Interaction	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
√	√	×	√	√

A1: Range of Resources

A1 is covered. Three model family sources; Four Literature API's; Six team members; Rented Blackwell GPU. Coordinating these resources clearly has to be part of the work. The plan in Chapter 3 assigns each task to a specific team member and tracks weeks/deliverables. That is why A1 is labeled as covered.

A2: Level of Interaction

A2 is covered. The system makes heavy use of the outside world. Every round of debating calls three external model API endpoints and three Literature API endpoints, and there is communication to/from the rented cloud GPU. Team communication occurs via the shared repository and weekly meeting. High degree of interactivity = why A2 is labeled as covered.

A3: Innovation

A3 is not covered. The framework combines existing concepts: multi-agent debate, retrieval, and trust weighting. It creates no new device and no new theory. The contribution is the combination of claim-level external evidence with a bounded in-debate trust state and a controlled wrong-majority protocol. That combination is useful and testable, but it is not a patentable or breakthrough innovation. A3 is therefore marked as not covered.

A4: Consequences for Society and Environment

A4 is covered. The work targets a social problem: unreliable and sycophantic answers from AI systems in scientific question answering. Even though the framework is a lab prototype rather than a production-deployed product, the consequences of the problem and of the solution were both considered during design. The environmental impact was assessed qualitatively: the pre-registered matrix is 1,067–1,280 GPU-h on a single rented GPU, the project does no training, and the models run as quantized checkpoints. Energy use was not metered, so no quantitative energy claim is made. These social and environmental considerations shaped the ethical constraints and the inference-only design documented in this chapter.

A5: Familiarity

A5 is covered. Both multi-agent debate/evidence verification were uncharted territory for our team. Our literature-review phase and our baseline study that used a step-by-step implementation plan were both structured around the lack of prior experience with those areas. Only after we had studied each component sufficiently could we begin implementing each one. Lack of prior-experience with many components = why A5 is labeled as "covered".

5.4.3 Program Outcomes

Table 5.7 maps FYDP I against the twelve program outcomes.

Table 5.7: Mapping with program outcomes (PO1–PO12).

PO1 Engineering knowledge	PO2 Problem analysis	PO3 Design of solutions	PO4 Investigation	PO5 Modern tool usage	PO6 The engineer and society
✓	✓	×	✓	×	×
PO7 Environment and sustainability	PO8 Ethics	PO9 Individual and teamwork	PO10 Communication	PO11 Project management and finance	PO12 Life-long learning
×	×	×	✓	✓	✓

The real-life problem in Chapter 1 and the work in this report address six of the twelve program outcomes: PO1, PO2, PO4, PO10, PO11, and PO12. Literature review and gap analysis (Chapter 2) cover PO4. PO2 is covered by the requirements work in Chapter 3. PO10 is addressed in the interim report and in the oral presentation. The timeline and budget cover PO11. The journal writing and the independent learning record cover PO12. The remaining six outcomes are scheduled for FYDP II and FYDP III and are marked as not covered in Table 5.7.

5.5 Summary

The chapter defined the requirements for the software, hardware, and communication layers and documented the economic, environmental, ethical, and social constraints that shaped design decisions. The budget analysis shows that the pre-registered matrix is 1,067–1,280 GPU-h, with a Core-run planning envelope of USD 420–580 all-in and a Full-run top-up of USD 900–1,200. The complex engineering mapping shows that six of the seven P attributes are covered, with P6 not covered because there are no external stakeholders. K1–K8 are all covered. A1–A2, A4, and A5 are covered, while A3 is not. For the program outcomes, FYDP I covers PO1, PO2, PO4, PO10, PO11, and PO12.

Chapter 6

Conclusion

6.1 Summary

This report examined whether evidence-grounded trust can reduce sycophantic collapse in multi-agent large language model debate. The problem is concrete: a confident but wrong majority can pressure a correct minority agent into abandoning the correct answer, and a final majority vote then presents the result as consensus. The proposed response replaces the equal-weight vote with a trust vector that is updated during the debate from claim-level scientific evidence. Each agent's response is decomposed into claims, each claim is routed to a domain-specific source pair, and a verifier assigns supported, contradicted, or unverifiable. Supported and contradicted claims update the agent's weight through logit accumulation, one softmax, and a simplex projection. The final answer is a trust-weighted plurality.

The project produced three artifacts. The first is an implemented pipeline: a three-agent, three-round debate loop; seeded option rendering; an answer parser and scorer; claim decomposition; claim-domain retrieval; a verifier interface; the bounded trust operator; weighted aggregation; injection; and a resumable record writer. The second is a validation harness: a 514-test suite, a 13-check P0 gate, a mock end-to-end path, and the Gate 0 transcript. The third is a pre-registered evaluation plan: four executable arms across three datasets and two seeds, 24 runs, 1,280 GPU-h under the conservative cap, and a six-arm Full target whose placebo and oracle controls are specified under WP-18.

6.2 What Was and Was Not Shown

The report shows that the base debate loop runs end to end. Gate 0 completed 10 of 10 debates on a ten-question GPQA slice, produced 90 of 90 non-empty positions, and averaged 370.8 seconds per debate with a standard deviation of 21.5 seconds. The report also shows that the P0 hardening work is present: options are rendered into every multiple-choice prompt with a seeded permutation, the parser handles explicit answer spans and scientific notation, the trust operator satisfies its floor and sum constraints over 30,000

random draws for one, three, and five agents, the eight endpoint definitions run on schema-valid records, the dataset revisions are pinned, claims route by domain, the manifest guard refuses a mismatched resume, and the matrix is cut to a pre-registered 24-row plan.

Those results are wiring and validation evidence. They do not show that trust weighting reduces collapse, improves accuracy, or preserves a correct minority better than majority voting. The comparative matrix has not been run. The verifier has not been human-audited. The final stack has not been smoke-tested on the 96GB card. No human evaluation has been conducted. The report therefore makes no efficacy claim, and the abstract, introduction, and conclusion are written to that boundary.

6.3 Limitations

The most important limitation is verifier quality. The default verifier is lexical, and lexical overlap is not entailment. It can mistake a shared term for support, miss a negation, and misread a numeric claim. The cross-encoder is optional, and the NLI and LLM-judge paths are interfaces rather than validated components. A trust signal built on a weak verifier can amplify a confident wrong majority instead of correcting it. The pre-registration treats verifier accuracy as a moderator for this reason, but the audit that would measure it is not complete.

The second limitation is the evidence layer's coverage. The live PubMed, arXiv, Semantic Scholar, and OpenAlex calls are not exercised by the test suite. Routing and deduplication are tested, but rate limits, source outages, and domain coverage are operational unknowns. A claim with no retrieved passage becomes unverifiable and does not move trust, so sparse domains weaken the mechanism. The report states this as a boundary rather than hiding it.

The third limitation is the experimental scope. Gate 0 covers ten GPQA questions from one dataset, one development stack, and one three-model panel. The matrix is planned for GPQA-Main, MMLU-Pro STEM, and HLE, but it has not run. The injection protocol is a controlled stress test, not natural sycophancy. The debate is fixed at three rounds and three agents, and the model families are three specific open checkpoints. The results, when they exist, will be conditional on those choices.

The fourth limitation is implementation timing. The trust timing is fixed: the debate loop performs a closing pass on the final positions and exposes the final-position trust vector. The runner-side record adapter copies the manifest hash, the final-answer fields, and the final-position trust vector into every record. The six-arm target is documented and pre-registered, but the placebo and oracle arms depend on WP-18. The final stack is pinned by revision and weight total, not by a successful serving run. These are engineering gaps, not evidence of failure, but they bound what the current implementation can support.

6.4 Future Work

The next step is the final-stack smoke test on the 96GB card, the WP-18 placebo/oracle slice, the 24-item calibration batch, and the verifier human audit. The runner adapter already stamps every record with the manifest hash and the final-answer and trust fields. These steps are prerequisites for a clean run.

The verifier audit is the scientific prerequisite. A set of 150–200 claim-passage pairs, labelled by two annotators, should be used to report balanced accuracy, precision, recall, coverage, and agreement. If the lexical verifier falls below the pre-registered threshold, the cross-encoder or an LLM judge should be substituted and re-audited before the matrix. A matrix run on a verifier that has not passed the audit would produce a number, but not a trustworthy one.

After the audit, the executable matrix can run: four arms, three datasets, two seeds, and the pre-registered analysis. The oracle arm belongs to the six-arm Full target under WP-18; the Core run is the four-arm prefix and does not include it. The primary contrast is B8 versus B8-no-trust. The co-primary endpoint is correct-minority retention. The secondary endpoints are appropriate revision, trust AUROC, trust Brier, verifier accuracy, and citation faithfulness. The six-arm target adds the placebo and oracle controls. A negative or null result is reportable under the pre-registration; it would bound the conditions under which evidence-grounded trust helps, which is itself useful.

The final-stack smoke test, the N-scaling and α/β ablations, and the human evaluation follow. The human evaluation should use blinded binary judgments rather than a seven-point Likert scale as the primary endpoint. The longer-term goal is a reproducible package with code, configuration, prompts, seeds, dataset identifiers, and the pre-registration, so that another group can repeat the matrix and inspect the evidence behind each answer.

The project has not shown that trust-calibrated deliberation works. It has shown that the mechanism can be built, tested, bounded, and pre-registered, and it has separated the evidence that exists from the evidence that does not. That separation is the condition for a credible test, and the test is the next step.

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